

Crisis Shield

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1. Project Title

CascadeWatch: Simulating Cascading Infrastructure Failures During Disasters

2. Primary Domain

ML / AI + Simulation (with Geospatial Data and Interactive Visualization)

3. Problem Statement

When a disaster strikes, damage rarely stops at the point of impact. A flood does not just submerge roads — it knocks out power substations, which disables hospitals, which overwhelms emergency services. These chain reactions, known as cascading failures, are responsible for a significant share of disaster-related casualties and economic damage.

So for our project we will build a models that shows how failures propagate across interconnected infrastructure in real time, provides help to planners and responders.

4. Project Description

Scope & Direction

CascadeWatch builds a graph-based simulation of a city's infrastructure network. Nodes represent components such as power substations, hospitals, water treatment plants, road junctions and communication towers. Edges represent operational dependencies. When a disaster event is triggered, a propagation model predicts which nodes fail, in what order, and with what probability.

Novelty

- Multi-layer infrastructure dependency modeling across power, water, transport and health
- Real geospatial data for an Indian city — not synthetic or hypothetical
- Animated, time-stepped visualization of cascading failures unfolding in real time
- User-controlled disaster parameters enabling scenario exploration and comparison

5. Datasets

- OpenStreetMap via Geofabrik / Overpass API — roads, power lines, hospitals (free, open license)
- Open Infrastructure Map — global power grid data (open license)
- data.gov.in — National Hospital Directory with geo-coordinates (free)
- EM-DAT (emdat.be) — historical disaster event records for India (free for academic use)

Data will be scoped to one major Indian layout to ensure completeness.

6. Technology Stack

Component	Tool / Technology
Data Sources	OpenStreetMap (roads, power, hospitals), data.gov.in, EM-DAT
Graph Engine	NetworkX (Python) or PyTorch Geometric
Simulation	Agent-Based Propagation Model
ML Layer	Failure Probability Classifier (Random Forest / GNN)
Visualization	Python Dash + Folium map, Plotly charts

7. Final Deliverables

- Working simulation dashboard with interactive map and animated failure cascade
- 3 fully tested disaster scenarios (flood, earthquake, cyclone) on a real Indian layout
- Impact severity report generator with timeline of failures
- Project report and final presentation